

A comprehensive investigation of the associations between tensor-based morphometry indices and executive functions, memory, language, and visuospatial abilities in patients in the Alzheimer's disease continuum

Sahba Azadikhah Jahromi^a, Aram Parhizkar^b, Mahtab Mohammadi^c, Danial Kazemi^d,
 Mohammad Hassan Tajik^e, Maryam Nazari^f, Maryam Bemanalizadeh^{g,h},
 Seyed Mohammad Amin Alavi^{i,*}, Alzheimer's Disease Neuroimaging Initiative

^a School of Mechanical Engineering, Iran University of Science and Technology (IUST), Tehran, Iran

^b Department of Pharmaceutical and Pharmacological Sciences, University of Padova, Italy

^c Department of Psychology, Faculty of Psychology and Educational Sciences, Central Tehran Branch, Islamic Azad University, Tehran, Iran

^d Student Research Committee, Isfahan University of Medical Sciences, Hezar Jerib Street, Isfahan, Iran

^e Department of Anatomy, Tehran University of Medical Sciences, Tehran, Iran

^f Medical Branch of Islamic Azad University of Tehran (IAUTMU), Tehran, Iran

^g NeuroTRACT Association, Students' Scientific Research Center, Tehran University of Medical Sciences, Tehran, Iran

^h Research Institute for Primordial Prevention of Non-Communicable Disease, Isfahan University of Medical Sciences, Isfahan, Iran

ⁱ Faculty of Medicine, Ahvaz Jundishapur University of Medical Sciences, Ahvaz, Iran

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ABSTRACT

Objectives: Based on the literature, tensor-based morphometry (TBM) parameters were related to neurocognitive functions such as memory, learning, language ability, and executive functions. The present study aims to evaluate the associations between TBM indices with executive functions, memory, language, and visuospatial abilities and the value of TBM in the clinical diagnosis of Alzheimer's disease (AD) among individuals with Alzheimer's disease continuum and mild cognitive impairment (MCI) from Alzheimer's Disease Neuroimaging Initiative (ADNI).

Methods: The authors used ADNI-memory (ADNI-MEM), ADNI-executive functions (ADNI-EF), ADNI-language (ADNI-LAN), and ADNI-visuospatial (ADNI-VS) composite scores. TBM parameters, including measure 1, which represents average within a statistically defined region-of-interest inside the temporal lobes and measure 2 which indicates average within an anatomically defined region-of-interest including bilateral temporal lobes were utilized in the current study. Statistical analysis was performed using IBM SPSS Statistics version 26, and Pearson's correlation, Bonferroni's correction, and multiple linear regression were utilized for data analysis. $P < 0.05$ was considered statistically significant.

Results: After screening 800 participants, 270 (151 men, 119 women) were selected for a study with TBM scores and cognition-related assessments at 6, 12, and 24 months. Groups included healthy controls ($n=53$), MCI ($n=158$), and Alzheimer's Disease (AD) ($n=59$). TBM indices correlated with cognitive scores in MCI and AD groups but not healthy controls. Changes in TBM indices and cognitive scores were significantly correlated in MCI and AD groups over 24 months. TBM indices were weak predictors of cognitive decline at all time points.

Conclusions: TBM can help physicians diagnose MCI and AD early. However, TBM could not strongly predict cognitive functions decline at all time points.

* Correspondence to: Faculty of medicine, Ahvaz Jundishapur University of Medical Sciences, Golestan Blvd., Ahvaz, Khuzestan 6135715794, Iran.

E-mail address: sma.alavi94@gmail.com (S.M.A. Alavi).

¹ ORCID: 0000-0001-7885-6691

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1. Introduction

Dementia is a widespread condition with worldwide cases anticipated to increase from 56.4 million cases in 2019 to 152.8 million in 2050 [1]. The most common cause of dementia is Alzheimer's disease (AD) which has become a global health issue owing to the aging of the population, with the number of afflicted patients estimated to triple to exceed 13.4 million in the USA by the year 2050 [2]. AD pathogenesis is accepted to start decades before the onset of the clinical symptoms, mainly manifesting with cognitive decline [3]. Recent memory impairment is an early sign of AD, followed by a gradual decrease in other domains of neurocognition, including orientation, attention, language, executive functions, visuospatial abilities, and behavioral and emotional changes [4].

Several diagnostic modalities for AD have been explored in the literature, each possessing distinct advantages and disadvantages. Conventional diagnostic approaches encompass clinical assessments, evaluations of medical history, and cognitive testing. Nonetheless, reliance solely on these methods may lead to delays in the diagnosis of AD due to inherent subjectivity and variability in the interpretation of clinical data and cognitive evaluations. The use of biomarkers from cerebrospinal fluid (CSF), notably tau proteins and beta-amyloid, facilitates early diagnosis but necessitates invasive lumbar puncture, expensive analysis, and advanced equipment. Additionally, imaging modalities such as magnetic resonance imaging (MRI), positron emission tomography (PET), and single-photon emission computed tomography (SPECT) serve as valuable tools in AD diagnosis. However, these approaches can be cost-prohibitive and demand specialized equipment and trained staff, and they may also require the use of radioactive tracers, posing potential risks and logistical challenges [5].

Neuroimaging modalities, particularly magnetic resonance imaging (MRI), is recommended for diagnosis and follow-ups of individuals with AD [6]. Tensor-based morphometry (TBM), an automated brain mapping method of structural MRI, is an effective method for 3D mapping of brain degeneration in AD cases. TBM uses non-linear deformation field gradients to identify regional structural variations [7]. TBM could indicate volumetric changes across the brain. It is a valuable method for both cross-sectional and longitudinal studies [8]. Because of its high sensitivity, TBM is considered an appropriate tool for showcasing the neuroanatomical changes through clinical trials and large-scale studies [9]. Previous literature indicated that multi-template TBM is an effective method for detecting the initial phases of dementia [10,11], which is a fully automated [10] and considered as a non-invasive modality.

Previous investigations have found TBM as a helpful method for anticipating the future decline of neurocognitive performance and memory, predicting conversion of mild cognitive impairment (MCI) to AD, and tracking AD progression [12,13]. However, correlations between TBM indices and some other brain functions were discussed in much fewer studies. Based on these studies, TBM parameters were related to neurocognitive functions such as memory, learning, proverb interpretation, and executive functions [14–17]. A study by Barrows et al. examined 15 patients with behavioral variant frontotemporal dementia (bvFTD) and 16 patients with early-onset AD (eAD) using the Clock Drawing Test (CDT). TBM analysis revealed that the accuracy of hand placement in the task was associated with the volume of the right frontal lobe. These findings indicate that reduced performance on hand placement tasks is linked to the involvement of the right dorsolateral frontal-parietal network, which is crucial for executive control and is particularly affected in eAD [14]. Kaiser et al. conducted a study involving 11 patients with bvFTD and 10 patients with AD, assessing them through a series of executive function and semantic knowledge tests, along with TBM. Their findings revealed that, even after controlling for the type of dementia, TBM analyses showed significant correlations between impaired proverb interpretation and the anterior temporal lobe, with a stronger effect observed on the left side compared to the right [15]. Tuokkola et al. investigated deep gray matter (DGM)

volume changes in three groups: controls ($n = 58$), individuals with MCI ($n = 38$), and patients with AD ($n = 58$) using TBM. Their study found that the AD group exhibited the most significant TBM volume changes across all DGM regions. The differences in volume changes were more pronounced between the control and AD groups compared to other group pairs. In the AD group, notable volume reductions in the right thalamus were significantly associated with impairments in episodic memory, learning, and semantic processing. Additionally, changes in the bilateral caudate nucleus were significantly or nearly significantly correlated with episodic memory and semantic processing. In contrast, few significant associations were found in the control and MCI groups [16]. Lin et al. utilized the Reliable Change Index (RCI) to identify participants who exhibited significant declines in Alzheimer's Disease Neuroimaging Initiative-memory (ADNI-Mem; $N = 450$) and executive functioning (ADNI-EF; $N = 456$) factor scores. The analysis revealed that reliable declines in ADNI-Mem scores were associated with atrophy in the medial temporal cortex, limbic cortex, temporal lobe, and parts of the parietal lobe. A similar pattern of atrophy was observed for those with reliable declines in ADNI-EF scores, although in this case, the atrophy extended into the frontal lobe and was more extensive and pronounced [17]. In earlier studies, different TBM indicators such as baseline temporal lobe atrophy, right frontal volume, and deep gray matter atrophy were correlated with impairment of neurocognitive functions [13,14,16].

Prior studies mainly focused on associations of a few aspects of cognitive performance impairment of AD patients and TBM indicators. Thus, the present study was carried out to evaluate the associations between TBM indices with executive functions, memory, language, and visuospatial abilities among individuals with Alzheimer's disease continuum from ADNI database [18].

2. Materials and methods

2.1. Participants and design

The data utilized for this article were acquired from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database, which can be found at adni.loni.ucla.edu. The ADNI was initiated in 2003 through a collaboration between the National Institute on Aging (NIA), the National Institute of Biomedical Imaging and Bioengineering (NIBIB), the Food and Drug Administration (FDA), private pharmaceutical companies, and non-profit organizations. It was established as a public-private partnership with a funding of \$60 million over a period of 5 years. The main objective of ADNI has been to evaluate the feasibility of integrating serial MRI, PET, and other biological markers to quantify the advancement of MCI and early AD. The objective of identifying sensitive and specific markers of early AD progression is to assist researchers and physicians in the development of novel treatments and the evaluation of their efficacy, while also reducing the duration and expenses associated with clinical trials. The principal investigator for this effort is Michael W. Weiner, MD, from the VA Medical Center and the University of California-San Francisco. To access the latest information, please visit the website www.adni-info.org.

We included all individuals with healthy control (HC), MCI, and AD diagnosis from the ADNI database, for whom data regarding TBM indices and executive functions, memory, language, and visuospatial composite scores were available at baseline, 6 months, 12 months, and 24 months timepoints. Moreover, demographic characteristics were further added for analysis. Eventually, 270 participants were determined to be eligible for the investigation, comprising 53 HC, 158 MCI, and 59 AD individuals.

2.2. Cognitive assessments

The authors used ADNI-MEM and ADNI-EF composite scores, which are psychometrically advanced measures of memory and executive

functioning, respectively, to identify individuals with significant alterations in cognitive abilities. ADNI-MEM and ADNI-EF were described by Crane et al. (2012) and Gibbons et al. (2012), respectively [19,20]. The ADNI-MEM composite score is calculated based on the recorded scores obtained from the administration of several cognitive assessment tools, including the Mini-Mental State Examination (MMSE), Alzheimer's Disease Assessment Scale - Cognitive Sub-scale (ADAS-Cog), Auditory Verbal Learning Test (RAVLT), and Logical Memory [21–24]. In a similar way, the factor scores for ADNI-EF are obtained by utilizing observed scores on various cognitive tests, including Digit Symbol Coding and Digit Span, Trail Making Test A and B, category fluency (animal and vegetable), and the clock drawing test [25–29]. ADNI-LAN and ADNI-VS were also utilized to assess the language and visuospatial functioning of individuals [30]. The Boston naming test, MMSE, ADAS-Cog, and Montreal Cognitive Assessment (MoCA) were administered to evaluate participants' language composite (ADNI-LAN) [31,32]. Finally, 5 Clock Drawing items, MMSE, and ADAS-Cog were utilized to develop ADNI-VS [33]. ADNI-MEM, ADNI-EF, ADNI-LAN, and ADNI-VS composite scores for each individual are reported in the UWN-PSYCHSUM file, which can be found on the ADNI website.

2.3. Neuroimaging

The study involved measuring brain volumes using MRI scans from various scanners, with quality control to exclude artifacts. The images were preprocessed by aligning them to a standard brain space (ICBM-53) using a 9-parameter transformation, followed by resampling to achieve uniform voxel sizes. TBM was employed to analyze brain structural changes, focusing on brain atrophy and using techniques like skull stripping and brain masking to improve accuracy. Longitudinal TBM specifically tracked changes over time by aligning follow-up scans with baseline scans. Regions of interest (ROIs) in the brain were measured and compared for atrophy across different brain regions. Fig. 1 illustrates a detailed description of neuroimaging and TBM. Moreover, a complete description of neuroimaging and TBM can be found on the ADNI website (<http://adni.loni.usc.edu>). It should be mentioned that none of the scans were excluded in the current study to ensure that the methods are unbiased.

In this study, we employed two parameters for TBM. Measure one is a quantitative summary of the total atrophy in the temporal lobes, calculated as the average within a statistically defined region-of-

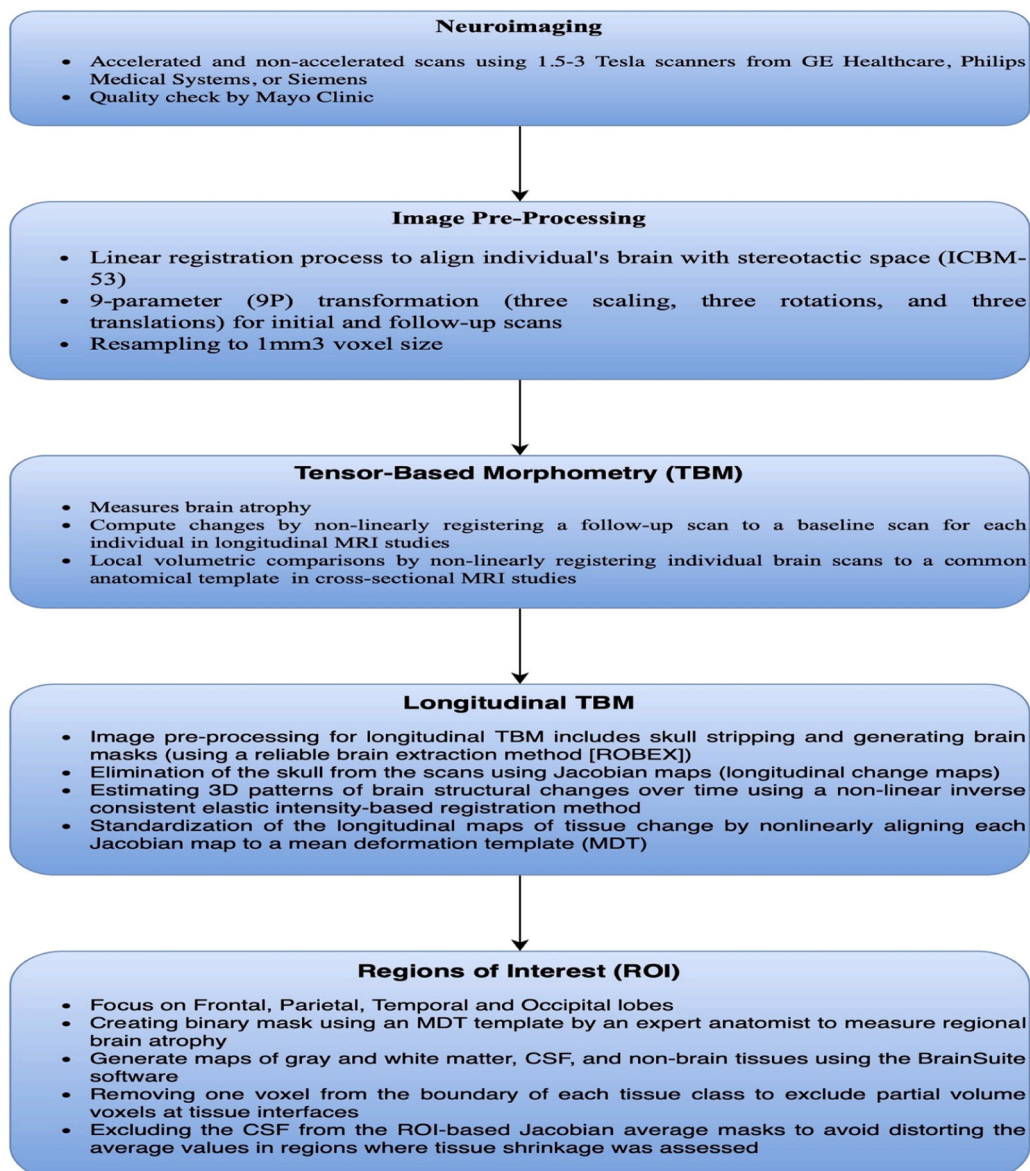


Fig. 1. The process of neuroimaging and Tensor-Based Morphometry (TBM) used to assess brain structure.

interest. Measure two is a quantitative summary of the total atrophy in the temporal lobes, calculated as the average within an anatomically defined region-of-interest that includes temporal lobes.

3. Statistical analyses

For each diagnostic group (AD, MCI, and normal aging), descriptive statistics and frequencies were generated for participant demographics, clinical features, neuroimaging, and neurocognitive tests, which were associated with brain atrophy rates [14]. A summary of these evaluations is provided below.

While the differences in sex and handedness were evaluated by the Chi-square test, the analysis of variance (ANOVA) was employed to compare age differences across groups. This analysis was modified by the Bonferroni post-hoc test for multiple comparisons. To assess educational differences, the non-parametric Kruskal-Wallis test was applied. The statistical results of the TBM indices and cognitive tests pertaining to memory, language, executive functions, and visuospatial abilities, were further analyzed using the analysis of variance (ANOVA), with Tukey's corrections for pairwise comparisons. Multiple linear regression analyses were performed to measure associations between differences in composite scores and differences of domain-specific cognitive functions from baseline in TBM parameters. The effect sizes of three groups on the cognitive function findings were also measured by Cohen's d. Except for Cohen's d, which was obtained from the Social Science Statistics Website, all statistical analyses in this study were conducted using IBM SPSS ver. 26 (IBM Corp, Armonk, NY, USA). $P < 0.05$ was considered statistically significant.

4. Results

After screening 800 participants for this study, we selected 270 (151 men and 119 women) individuals with both TBM scores and cognition-related assessments at three time points (6, 12, and 24 months after the baseline) and put them into three groups: HC (n=53), MCI (n=158), and AD (n=59). The mean ages of the normal, MCI, and AD were 76.1 ± 6.1 , 73.9 ± 7.61 , and 76.6 ± 7.42 , respectively.

5. Demographic characteristics

The demographic characteristics and the results of the statistical analyses are shown in Table 1. The mean years of education for the

Table 1
Demographic characteristics of the included population.

Demographic data		HC (n=53)	MCI (n=158)	AD (n=59)	<i>p</i>
Gender	Male	28 (52.8)	93 (58.9)	30 (50.8)	0.503
	Female	25 (47.2)	65 (41.1)	29 (49.2)	
Age		76.1 ± 6.10	73.9 ± 7.61	76.6 ± 7.42	0.027
Years of Education		14.6 ± 3.28	15.4 ± 2.94	14.1 ± 2.99	0.017
Ethnicity	Hispanic or Latino	2 (3.8)	4 (2.5)	0	0.368
	Not Hispanic or Latino	51 (96.2)	154 (97.5)	59 (100)	
Race	Asian	0	2 (1.3)	0	0.003
	Native Hawaiian or Other Pacific Islander	0	1 (0.6)	0	
	Black or African American	10 (18.9)	6 (3.8)	1 (1.7)	
	White	43 (81.1)	149 (94.3)	58 (98.3)	

* Significant p values appeared in boldface.

Abbreviations: HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer disease;

normal, MCI, and AD groups were 14.6 ± 3.28 , 15.4 ± 2.94 , and 14.1 ± 2.99 , respectively. There were no significant differences in the distribution of gender and ethnicity between the groups ($P=0.503$ and $P=0.368$). However, all groups differed significantly in terms of racial categories, age, and years of education ($P=0.003$, $P=0.027$, and $P=0.017$).

5.1. Correlation between TBM indices and composite scores of domain-specific cognitive functions across groups

The correlation between TBM indices and the four composite scores of MEM, EF, LAN, and VS across the three groups (i.e., HC, MCI, and AD) has been shown in Table 2. In general, TBM indices were not correlated with the composite scores of MEM, EF, LAN, and VS in the HC group through the 24 months' follow-up, except for the significant correlation between measure 2 and the VS composite score at month 6 ($P=0.025$). However, in the MCI group, there was a significant correlation across time between the TBM indices and the composite scores of MEM, EF, LAN, and VS, except for the correlations between the TBM indices and the VS composite score at months 6 and 12, which were not significant (P for measure 1 = 0.881 and 0.113 , respectively, and P for measure 2 = 0.365 and 0.453 , respectively). Although the correlations between TBM indices and all four composite scores (MEM, EF, LAN, VS) were significant at 24 months in the AD group (P for measure 1 = <0.0001 , 0.005 , 0.002 , <0.0001 , respectively, and P for measure 2 = 0.023 , 0.013 , 0.032 , 0.006 , respectively), we found no significant correlation between these measures earlier at 6 and 12 months, except for the correlation between measure 1 and LAN and measure 2 and EF composite score at month 12 ($P=0.036$ and $P=0.031$, respectively).

5.2. Correlation between variations in TBM indices and changes in composite scores of domain-specific cognitive functions from baseline across groups

Table 3 presents the correlation between changes in TBM indices from baseline and the four composite scores—MEM, EF, LAN, and VS—across the three groups (HC, MCI, and AD). Overall, within the HC group, changes in TBM indices over the 24-month follow-up period generally did not correlate with changes in the composite scores of MEM, EF, LAN, and VS. However, two exceptions were noted: a significant negative correlation between changes in measure 2 and the VS composite score after six months ($P=0.022$), and between changes in measure 1 and the MEM composite score after 24 months ($P=0.048$).

In the MCI group, significant negative correlations were observed between the MEM and LAN composite scores and measure 1 after 12 months ($P<0.001$ and $P=0.031$, respectively), as well as between measure 2 and these scores after 24 months ($P<0.001$ and $P=0.003$, respectively). Notably, all composite scores showed a significant negative correlation with measure 1 after the 24-month follow-up period ($P<0.001$, $P<0.001$, $P<0.001$, and $P=0.004$, respectively).

In AD patients, a significant correlation was observed between the EF composite score and measure 1 after six months ($P=0.003$). Additionally, after 24 months, the MEM composite score was significantly correlated with measure 2 ($P=0.033$). Moreover, after 24 months, all composite scores demonstrated significant negative correlations with measure 1 ($P=0.007$, $P<0.001$, $P=0.004$, and $P=0.013$, respectively).

5.3. Association between variations in TBM indices and changes in composite scores of domain-specific cognitive functions from baseline

The association between TBM indices changes and changes in composite scores of domain-specific cognitive functions from baseline and their results, including F-value and P-value, are shown in Table 4. A multiple linear regression analysis was conducted at three distinct time points—6, 12, and 24 months post-baseline while adjusting for gender, age, and years of education. Neither measure 1 nor measure 2 could not

Table 2

The correlation between TBM indices and composite scores of domain-specific cognitive functions across groups.

Timepoint	Diagnosis	Measure	MEM (p value and correlation coefficient)	EF (p value and correlation coefficient)	LAN (p value and correlation coefficient)	VS (p value and correlation coefficient)
6 months	HC	Measure 1	0.544, 0.085	0.225, 0.170	0.786, 0.038	0.498, 0.095
		Measure 2	0.403, 0.117	0.154, 0.198	0.437, 0.109	0.025 , 0.308
	MCI	Measure 1	<0.001 , 0.313	<0.001 , 0.259	0.001 , 0.259	0.881, −0.012
		Measure 2	0.022 , 0.182	0.016 , 0.192	0.029 , 0.174	0.365, 0.073
	AD	Measure 1	0.363, 0.121	0.280, 0.143	0.257, 0.150	0.746, −0.043
		Measure 2	0.955, 0.007	0.212, 0.165	0.242, 0.155	0.483, 0.093
	HC	Measure 1	0.698, 0.055	0.061, 0.259	0.193, 0.182	0.126, 0.213
		Measure 2	0.866, 0.024	0.366, 0.127	0.120, 0.216	0.101, 0.228
	MCI	Measure 1	<0.001 , 0.475	<0.001 , 0.339	<0.001 , 0.402	0.113, 0.127
		Measure 2	<0.001 , 0.289	0.007 , 0.213	0.002 , 0.240	0.453, 0.060
	AD	Measure 1	0.188, 0.174	0.128, 0.201	0.031*, 0.282	0.146, 0.192
		Measure 2	0.408, 0.110	0.036 , 0.274	0.206, 0.167	0.059, 0.247
12 months	HC	Measure 1	0.055, 0.265	0.218, 0.172	0.952, −0.008	0.942, 0.010
		Measure 2	0.800, 0.036	0.608, 0.072	0.116, −0.218	0.387, −0.121
	MCI	Measure 1	<0.001 , 0.648	<0.001 , 0.485	<0.001 , 0.500	0.007 , 0.212
		Measure 2	<0.001 , 0.436	0.001 , 0.272	<0.001 , 0.317	0.049 , 0.157
	AD	Measure 1	<0.001 , 0.467	0.005 , 0.362	<0.001 , 0.449	0.002 , 0.400
		Measure 2	0.023 , 0.296	0.013 , 0.321	0.032 , 0.280	0.006 , 0.352
24 months	HC	Measure 1	0.055, 0.265	0.218, 0.172	0.952, −0.008	0.942, 0.010
		Measure 2	0.800, 0.036	0.608, 0.072	0.116, −0.218	0.387, −0.121
	MCI	Measure 1	<0.001 , 0.648	<0.001 , 0.485	<0.001 , 0.500	0.007 , 0.212
		Measure 2	<0.001 , 0.436	0.001 , 0.272	<0.001 , 0.317	0.049 , 0.157
	AD	Measure 1	<0.001 , 0.467	0.005 , 0.362	<0.001 , 0.449	0.002 , 0.400
		Measure 2	0.023 , 0.296	0.013 , 0.321	0.032 , 0.280	0.006 , 0.352

* Significant p values appeared in boldface.

Abbreviations: HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer disease; Mem, Memory; EF, Executive Function; Lan, Language; VS, Visuospatial; Measure 1, numerical summary of cumulative temporal lobe atrophy – average within a statistically defined region-of-interest inside the temporal lobes; Measure 2, numerical summary of cumulative temporal lobe atrophy – average within an anatomically defined region-of-interest including bilateral temporal lobes;

strongly predict the cognitive functions decline at all timepoints.

6. Discussion

The present investigation represents one of the most comprehensive MRI analyses conducted thus far, with the objective of evaluating the associations between indicators of TBM, measure 1 and measure 2, alongside cognitive domains such as executive functions, memory, language, and visuospatial skills, across three cohorts comprising HC, MCI, and AD sourced from the ADNI database over the time intervals of 6, 12, and 24 months.

The primary findings derived from this study are: 1) Employing the TBM analytical framework, we observed no statistically meaningful differences in the atrophy rates reflected in the numerical summaries obtained from scans performed at 6 and 12 months when comparing accelerated and nonaccelerated scans within each cohort. 2) No substantial differences in temporal lobe cerebral alterations were identified between accelerated and nonaccelerated TBM assessments at either 6 or 12 months across any of the participant groups (AD, MCI, or HC) or their diagnostic combinations, as established through voxel-based analyses. 3) Upon comparison of results among HC, MCI, and AD groups, a significant association was detected within the MCI and AD groups; conversely, no correlation with the TBM scores was noted within the HC group, with the exception of the 24-month mark. 4) After assessing potential relationships with changes in TBM indices from the baseline, it emerged that these associations were considerably more pronounced in

the MCI and AD groups when compared to the HC group. And 5) The regression analysis used in this examination was unable to robustly predict the decline in cognitive functions across all assessed timepoints.

Our investigation reveals that the atrophy of the temporal lobe, evaluated via TBM, emerges as the principal biomarker for neurodegenerative disorders, demonstrating superior capability in differentiating AD patients from HC in comparison to other biomarkers. This observation corroborates earlier research that recognized temporal lobe atrophy as a hallmark of AD. One of these studies by Whitwell et al. (2007) examined the trajectory of cerebral atrophy in individuals progressing from amnesic mild cognitive impairment (aMCI) to AD through a series of MRI evaluations. The findings indicated that gray matter degeneration initially concentrates in the medial temporal lobes and advances considerably by the point of AD diagnosis [34]. In another investigation by Hua et al. (2016), the outcomes revealed pronounced tissue alterations in cerebral atrophy among individuals with MCI and AD, with specific manifestations such as left temporal lobe atrophy in MCI and medial temporal alongside limbic atrophy in AD [35]. In 2021, Chauveau et al. explored the distinct impacts of aging and AD on medial temporal lobe (MTL) atrophy throughout the entire continuum of AD. Their findings indicated that certain subregions of the MTL exhibit heightened susceptibility to atrophy during the early, prodromal phases of AD [36]. The utilization of TBM signifies a sophisticated methodology that has compellingly illustrated the trajectories of atrophy progression linked with various neurodegenerative disorders, demonstrating heightened accuracy in identifying remarkable atrophy when

Table 3

The correlation between variations in TBM indices from baseline and changes in composite scores of domain-specific of cognitive functions from baseline across time and groups.

Timepoint	Diagnosis	Measure	MEM (p value and correlation coefficient)	EF (p value and correlation coefficient)	LAN (p value and correlation coefficient)	VS (p value and correlation coefficient)
6 months	HC	Measure 1	0.626, −0.071	0.236, −0.171	0.522, 0.093	0.100, −0.235
		Measure 2	0.890, −0.020	0.960, 0.007	0.961, 0.007	0.022 , −0.324
	MCI	Measure 1	0.099, −0.133	0.290, −0.085	0.367, −0.073	0.683, −0.033
		Measure 2	0.405, −0.067	0.999, 0.000	0.607, −0.042	0.757, 0.025
	AD	Measure 1	0.156, −0.192	0.003 , −0.384	0.442, −0.105	0.862, −0.024
		Measure 2	0.869, −0.023	0.137, −0.201	0.698, −0.053	0.564, 0.079
	HC	Measure 1	0.602, −0.076	0.123, −0.221	0.944, −0.010	0.259, −0.163
		Measure 2	0.672, 0.061	0.422, −0.116	0.816, −0.034	0.883, −0.021
	MCI	Measure 1	<0.001 , −0.312	0.185, −0.107	0.031 , −0.173	0.073, −0.144
		Measure 2	0.050, −0.158	0.708, 0.030	0.272, −0.089	0.996, 0.000
	AD	Measure 1	0.112, −0.214	0.107, −0.218	0.256, −0.154	0.524, −0.087
		Measure 2	0.448, −0.103	0.073, −0.242	0.302, 0.140	0.803, 0.034
12 months	HC	Measure 1	0.048 , −0.281	0.274, −0.158	0.882, 0.022	0.737, 0.049
		Measure 2	0.335, −0.139	0.126, −0.219	0.502, 0.097	0.154, 0.204
	MCI	Measure 1	<0.001 , −0.531	<0.001 , −0.285	<0.001 , −0.337	0.004 , −0.229
		Measure 2	<0.001 , −0.353	0.472, −0.058	0.003 , −0.238	0.251, −0.093
	AD	Measure 1	0.007 , −0.355	<0.001 , −0.451	0.004 , −0.383	0.013 , −0.330
		Measure 2	0.033 , −0.285	0.088, −0.230	0.704, −0.052	0.266, −0.151

* Significant p values appeared in boldface.

Abbreviations: HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer disease; Mem, Memory; EF, Executive Function; Lan, Language; VS, Visuospatial; Measure 1, numerical summary of cumulative temporal lobe atrophy – average within a statistically defined region-of-interest inside the temporal lobes; Measure 2, numerical summary of cumulative temporal lobe atrophy – average within an anatomically defined region-of-interest including bilateral temporal lobes;

Table 4

Association between variations in TBM indices and changes in composite scores of domain-specific cognitive functions from baseline.

Measure	HC (R ² , F, p value)	MCI (R ² , F, p value)	AD (R ² , F, p value)
M1-diff-6	0.154, 1.175, 0.366	0.071, 1.639, 0.129	0.322, 3.459, 0.004
M2-diff-6	0.212, 1.725, 0.127	0.022, 0.475, 0.851	0.250, 2.434, 0.031
M1-diff-12	0.074, 0.517, 0.817	0.180, 4.709, <0.001	0.257, 2.526, 0.026
M2-diff-12	0.050, 0.335, 0.934	0.044, 0.976, 0.451	0.318, 3.399, 0.005
M1-diff-24	0.272, 2.402, 0.035	0.320, 10.062, <0.001	0.421, 5.294, <0.001
M2-diff-24	0.189, 1.497, 0.193	0.169, 4.344, <0.001	0.318, 3.395, 0.005

* Significant p values appeared in boldface.

** Adjusted for gender, age, and years of education

Abbreviations: HC, healthy control; MCI, mild cognitive impairment; AD, Alzheimer disease; M1diff, difference of numerical summary of cumulative temporal lobe atrophy – average within a statistically defined region-of-interest inside the temporal lobes with baseline; M2diff, difference of numerical summary of cumulative temporal lobe atrophy – average within an anatomically defined region-of-interest including bilateral temporal lobes with baseline

juxtaposed with subtle localized alterations [37,38].

Furthermore, our research elucidated a significant association between atrophy metrics derived from TBM and composite indices of MEM, EF, LAN, and VS in patients diagnosed with AD and MCI after a 24-month period. Clinical markers indicating a decline in cognitive functions, such as the MMSE and clinical dementia rating (CDR) scores, exhibited a robust correlation with both atrophy and ventricular enlargement. The findings from this investigation align with those reported in prior studies, as elaborated in the following. Hua et al. (2008) conducted a study encompassing three cohorts: HC, MCI, and AD. The researchers evaluated participants' brain atrophy employing the aforementioned assessments. The presence of initial atrophy in the temporal lobe was associated with current cognitive functions (MEM and EF), prospective cognitive deterioration, and the emergence of AD in individuals with MCI within the subsequent year. Additionally, this atrophy functioned as a prognostic indicator for HC [13].

Leow et al. (2009) executed a study aimed at analyzing the rates of deterioration in brain structures among individuals with TBM, along with the associated biomarkers and cognitive capabilities. The investigation compared longitudinal alterations in brain structures over a 12-month period across three groups: HC, MCI, and AD. The results demonstrated cerebral degeneration in AD patients and a comparatively milder pattern of degenerative alterations in those with MCI. The atrophy rate in the temporal lobe among MCI individuals was correlated with scores in logical/verbal learning memory, CDR, and fluctuations in

MMSE scores. The data suggested that individuals with MCI who advanced to AD experienced a considerably accelerated progression of temporal lobe atrophy relative to those who did not transition to AD [39].

In another inquiry by Brambati et al. (2009), TBM was utilized to monitor gray matter degeneration in subjects diagnosed with either left or right temporal lobe variants of semantic dementia over the span of one year. The results indicated a decrease in gray matter volume within the temporal lobe, with the most significant impact occurring at the initial stage. In addition, while the examination chiefly concentrated on the temporal lobe, it also recognized lateralized variations, though these were less marked relative to the initial atrophy evident in both types of semantic dementia [40]. McEvoy et al. (2011) examined the rates of regional brain atrophy utilizing longitudinal imaging across diverse cognitive stages, including cognitively intact adults, individuals with MCI, and those with AD. The study found that TBM-derived atrophy metrics were significantly associated with cognitive decline, particularly within memory and executive functions [41].

While all preceding studies established correlations between atrophy and MEM and EF scores, the current study has identified relationships between TBM and MEM, EF, LAN, and VS, thereby enriching the existing literature by demonstrating that these correlations intensify at the 24-month interval, implying that longitudinal evaluations may yield a more precise representation of cognitive impairment progression.

7. Strengths and limitations

The study examined the relationship between TBM indices and multiple cognitive domains, including memory, executive functions, language, and visuospatial abilities, over a 24-month period. This broad approach provided a more holistic understanding of how TBM correlates with cognitive decline across different stages of AD and MCI. The application of TBM as an advanced neuroimaging method is a significant strength, as it enables detailed 3D mapping of brain atrophy. TBM's sensitivity in detecting regional brain changes provided valuable insights into the neuroanatomical progression of MCI and AD.

The sample size is one of the main limitations of the current study. The relatively small sample size reduces the statistical power of the analysis, which may limit the generalizability of the findings and increases the risk of Type II errors. Future studies with larger sample sizes will help to confirm the correlation between TBM indices and cognitive decline, providing more robust and reliable results.

Another limitation of the current study is the variability in the MRI acquisition. MRIs were obtained from patients using three different devices with two different field strengths (1.5 and 3 Tesla). This variation can introduce inconsistencies in image quality and contrast, potentially leading to bias in the analysis and interpretation of the TBM indices. The differences in magnetic field strength can affect the signal-to-noise ratio and the resolution of the images, which in turn may impact the accuracy of the measured indices. These factors could lead to variability in the results, complicating the comparison of data across patients and potentially affecting the study's conclusions. With the advancement in MRI technology and the global increase in the use of 7 Tesla MRI scanners, which offer higher resolution and better signal-to-noise ratios, future studies should consider utilizing these more powerful MRI systems. This would help minimize the variability caused by different devices and field strengths, leading to more consistent and reliable data.

8. Conclusion

This study adds to the growing body of research on TBM by providing a comprehensive analysis of its associations with cognitive decline in individuals across the Alzheimer's disease continuum, including those with MCI and HC. Unlike previous studies that focused on specific aspects of cognitive impairment, our research offers a broader

examination of the relationship between TBM indices and various cognitive domains, including memory, executive functions, language, and visuospatial abilities, over a 24-month period. The findings reinforce the utility of TBM in tracking brain atrophy and clinical diagnosis of MCI and AD patients.

However, while the study confirms TBM's effectiveness in identifying significant neuroanatomical changes, it also reveals limitations in its predictive power for early cognitive decline. This suggests that TBM, in its current form, may not fully capture the subtle, early-stage changes in cognitive functions, underscoring the need for methodological advancements. Looking forward, future research should focus on refining TBM techniques to improve sensitivity to early cognitive changes.

Ethics approval

The study was done following the guidelines outlined in the Declaration of Helsinki. The research activities received approval from the Institutional Review Boards (IRB) at the study sites involved.

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CRediT authorship contribution statement

Seyed Mohammad Amin Alavi: Writing – review & editing, Writing – original draft, Conceptualization. **Maryam Bemanalizadeh:** Writing – review & editing, Conceptualization. **Maryam Nazari:** Writing – original draft. **Sahba Azadikhah Jahromi:** Writing – review & editing, Writing – original draft. **Mohammad Hassan Tajik:** Writing – original draft. **Danial Kazemi:** Formal analysis. **Mahtab Mohammadi:** Writing – original draft. **Aram Parhizkar:** Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contribution

MB and SMAA contributed to the study conception and design. Statistical Analysis was carried out by DK. The first draft of the manuscript was written by SAJ, AP, MM, MHT, MN, and SMAA. All authors commented on previous versions of the manuscript. The final revision and editing of the manuscript were conducted by MB and SMAA. All authors read and approved the final manuscript.

Consent to participate

Participants provided written informed consent.

Consent for publication

Not applicable.

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